

Thomas Williams

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EDUCATION

University of Alabama at Birmingham

Bachelor's degree: Computer Science

GPA: 3.8 (Magna Cum Laude)

Minors: Business Administration, Information Systems

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, GDScript, JavaScript, HTML, CSS

Technologies, Frameworks, & Libraries: Godot, GitHub, PostgreSQL, Agentic AI, RAG (Retrieval-Augmented Generation), React, Node.js, Express.js, Selenium, OpenCV

TECHNICAL PROJECTS

Solo Gameplay Programmer and Researcher | A Rolling Start |

Jan 2026 – May 2026

- Built a **3D** based mobility game in **Godot** to simulate powered wheelchair joystick learning for children through a gamified experience.
- Designed core game systems such as player saves and local user accounts using the **Singleton design pattern**.
- Engineered a **logging system** to capture granular performance metrics for each player including **spatial movement** and collision frequency for quantitative analysis (pending IRB approval for clinical testing).
- Crafted **distinctive, thematic levels** with **dynamic designs** to keep engagement in exploration as users search through the world.
- Integrated a specialized **HID (Human Interface Device)** to simulate the feeling of a real powered wheelchair.

AI Integration and Backend Developer | Real News |

Jan 2026 – May 2026

- Collaborated with a team to create an **AI powered** news summary aggregator that transforms the way news is consumed by using **dynamic clustering** to group related articles and AI agents to create unique briefs.
- Engineered an asynchronous **multi-agent pipeline** to orchestrate data flow and decision making for final generation.
- Managed **end-to-end data lifecycle** to ensure integrity and consistency from ingestion to delivery.
- Built a **RAG (Retrieval-Augmented Generation) pipeline** to ground AI responses in verified article context **reducing hallucinations**.
- Created a **server-side AI usage system** that manages tiered quotas for anonymous and registered users via **IP hashes** and **local storage identity mapping**.

Developer | Drone Security System |

Feb 2025 - Apr 2025

- Collaborated with creating a Java-based **autonomous security system** for Tello drones, implementing an **asynchronous state machine** to manage seamless transitions between precision patrol patterns and **real-time biometric verification**.
- Integrated **OpenCV-powered facial recognition** and a custom dashboard to enable **real-time spatial simulation**, and **automated identity validation** against a database.

Solo Developer | AI Powered Desktop Voice Assistant |

Jul 2025 - Aug 2025

- Built a **Python-based** voice assistant utilizing **Mistral AI** for **natural language processing (NLP)** and command execution.
- Automated browser actions using the **Selenium** library, enabling **voice commands** to perform searches, navigate webpages, and manage browser tabs.

WORK EXPERIENCE

Waverly Advisors | IT Intern |

April 2025 – Present

- Troubleshooted and resolved complex issues including hardware, software, and network issues to maintain uptime for operations.
- Manage the manual extraction of new client data, reformatting the structure to match company standards before moving them to the network drive for final migration.
- Mapped out detailed network topologies of 10+ offices using Visio so my team could get a better understanding of each branch's network configuration, reducing network troubleshooting time.

Publix | Customer Service Staff, Grocery Clerk |

June 2023 – April 2025

- Managed refunds, exchanges, and resolving customer issues professionally.
- Assisted in store operations by coordinating with team members to improve work efficiency in the front end.
- Restocked shelves and maintained inventory organization to ensure a clean shopping environment.
- Balanced the safe and all registers to ensure funds were accounted for, and discrepancies were solved

Marco's Pizza | Insider |

May 2020 – Aug 2022